Cylinder Loop Weights and Planar Nesting: Sharper Finite-Core Errors, Integer-Grid Transfer, and Conditional Loop-Count Laws

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Abstract

We analyze the critical honeycomb cylinder partition with fugacity on every polygon separating two marked points, including contractible nests. Under the source's physical-energy trace-distance and small-singular-value hypotheses, a direct singular-value comparison improves its finite-core log-determinant error from $O(L\xi^{1/5})$ to $O(L\xi^{1/3})$. An explicit polynomial sampling argument removes a polynomial prefactor from the source's integer-grid continuation step, without introducing noninteger-dimensional probability measures. Conditional on the source's all-positive-fugacity exponent, we derive the normalized count law, its full large-deviation rate, and the corresponding planar normalization. The determinant, continuation, and tilting tools are classical; the increment is their quantified use at these particular source interfaces. The strip input, contour transport, infinite-operator estimates, and pressure-to-topology comparison are retained as dependencies, not independently certified global theorems. Contribution assessment: C3/5, a sharper source-specific finite-core error and explicit continuation and count-law consequences, conditional on the stated upstream analytic inputs.

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1 Model, normalizations, and proof boundaries

The source is OpenAI's September 26, 2026 manuscript *Cylinder loop weights and planar nesting* [1]. All references to its equations and lemmas below refer to the captured source, not to an asserted live branch state. Its honeycomb lattice is dual to triangular tiles of side one. Put

$$\rho = (2+\sqrt{2})^{-1/2}, \qquad \lambda = \pi/8, \quad \beta = \pi/4, \quad \gamma = 4/3, \qquad L = \gamma^{-1}\log N.$$

For even $N = 2m$, the cylinder period is $iN\sqrt{3}/2$; the marked points have displacement $-N/4 + iN\sqrt{3}/4$. A polygon is simple, unrooted and unoriented, and $|P|$ counts its vertices. A family consists of pairwise vertex-disjoint polygons, each separating the two markers on the cylinder compactified by its ends. Both essential polygons and contractible polygons enclosing exactly one marker can qualify. Winding polygons alone give a different partition.

Write

$$Z_N(\chi) = \sum_{k\ge 0} w_N(k)\chi^k, \qquad w_N(k) = \sum_{\substack{\mathcal{F} \text{ admissible} \\ |\mathcal{F}|=k}} \prod_{P\in\mathcal{F}} \rho^{|P|}. \tag{1}$$

Here $w_N(0) = 1$; no factor 2^k is hidden in $w_N(k)$. The finite cut joining the markers bounds the number of disjoint separating polygons, so the sum in k is a polynomial. Finiteness of each coefficient on the longitudinally infinite cylinder is part of the source's fixed-width transfer analysis. For $\chi > 0$,

$$\mathbb{P}_{N,\chi}\{K = k\} = \frac{w_N(k)\chi^k}{Z_N(\chi)} \tag{2}$$

is the normalized count law. Complex contour sectors used to evaluate Z_N are not probability measures.

The source's all-fugacity assertion is

$$\lim_{\substack{N\to\infty \\ N \text{ even}}} \frac{\log Z_N(\chi)}{\log N} = \tau(\chi), \qquad \chi > 0, \tag{3}$$

where

$$\tau(\chi) = \begin{cases} \dfrac{1}{6} - \dfrac{2}{3\pi^2}\arccos^2(\chi/2), & 0 < \chi \le 2, \\ \dfrac{1}{6} + \dfrac{2}{3\pi^2}\operatorname{arcosh}^2(\chi/2), & \chi \ge 2. \end{cases} \tag{4}$$

It has continuous extension $\tau(0) = 0$. We use (3) explicitly as a hypothesis in the probabilistic consequences, rather than treating fresh reading as independent certification of every step of its proof.

1.1 Architecture and retained obligations

Source module	Scientific role and boundary retained here
`02_cylinder`	Finite local algebra, vacuum state, three-label contour representation, confluence, and relative determinant normalization. Fixed-width transfer decay is not a uniform-in-width mixing estimate.
`03_gas_mean`	Mean cancellation, compensated complex energy, physical Hilbert norm, gap/count estimates, determinant limit, smooth finite core and Schur complement. Our main refinement changes its last determinant estimate.
`03_reference`	Actual Legendre ensembles, linear-statistic estimates and compensated quadratic integration. Empirical centering is not replaced by a Gaussian law.
`04_gas_fluctuations`	Finite-dimensional particle contour transport and Jacobian cancellation, using a complex-symmetric Hessian and Hermitian norm estimates separately.
`04b_interpolation`	Holomorphic approximants to actual integer-particle integrals, with finite polynomial and endpoint expansions; no fractional particle law.
`04c_exponent`	Imbalance contour, alias control, complex upper bound, superharmonic anchor at $Z_N(0) = 1$, then coefficient positivity.
`05_pressure`	Separate annular and open-strip pressure problems, essential-loop and diameter removal, comparison with two planar nests, then length estimates.
`06_all_count`	Uniform large-fugacity upper bound and sub-Gaussian bounds on the *unnormalized* coefficients $w_N(k)$.

The strip-crossing companion is a genuine external input. Its boundary winding identity and $N^{-1/4}$ crossing law are not proved in this report. The source's denominator contributes $5\log N/24$; the interior gas determinant and reference factors contribute the remaining terms leading to $7/8 - 1/2 - 5/24 = 1/6$. A finite Gram calculation cannot supply these limiting statements.

2 A sharper physical finite-core determinant bound

2.1 The exact source interface

The source uses the complexification $\mathcal{X}$ of a real reference-energy Hilbert space. For primitive sine coefficients b_n its norm is

$$h_0 = \pi^2 \sum_{n\geq 1} n|b_n|^2.$$

The compensated interaction is represented by $T = I + j^*\mathscr{K}j$, with adjoints taken in this physical norm. Its bilinear energy matrix is complex symmetric in a real basis; it need not be Hermitian. The relevant source conclusions are, with constants independent of L,

$$\|T - I\|_1 \leq CL, \quad \|T\| \leq C, \quad \operatorname{Re} T \geq cL^{-2}I, \quad n_T(t) \leq CL\sqrt{t} \quad (0 < t < t_0 < 1). \tag{5}$$

The same small-value bound is available for orthogonal compressions. Here $n_T(t)$ counts singular values below t, with multiplicity; the infinitely repeated essential value 1 is not counted. For a real orthogonal finite-core projection P, the source constructs

$$B = PTP|_{P\mathcal{X}} \oplus I_{(I-P)\mathcal{X}}, \qquad \|T - B\|_1 \leq \xi L, \qquad \operatorname{rank} P = O_\xi(L). \tag{6}$$

No effective uniform rank constant in ξ is asserted. Changing to nonorthogonal coordinate vectors requires the corresponding Gram metric; Euclidean singular values of their raw matrix are not the quantities in (5).

Lemma 2.1 (Singular-value variation)**.** *For complex square matrices A, B of the same size, their singular values in decreasing order satisfy*

$$\sum_i |s_i(A) - s_i(B)| \le \|A - B\|_1 .$$

Proof. This is the trace-norm case of the classical singular-value variation inequality [2]. For clarity, it follows from the Hermitian dilations $H(A) = \left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$. For a Hermitian affine path $H(t)$, its ordered eigenvalues are absolutely continuous. At almost every t their derivatives are diagonal entries of $H'(t)$ in an orthonormal eigenbasis (diagonalize the derivative inside repeated eigenspaces). For any orthonormal basis, $\sum_i |\langle e_i, H'e_i\rangle| \le \operatorname{Tr}|H'|$, by taking the signed diagonal contraction and trace duality. Integration gives $\sum_i |\lambda_i(H(1)) - \lambda_i(H(0))| \le \|H(1) - H(0)\|_1$. The dilation eigenvalues are the ordered s_i and $-s_i$; both sides double, giving the assertion. Approximation, or the same almost-everywhere argument, includes zero singular values and crossings. □

Theorem 2.2 (Clipped logarithms without squaring)**.** *Let A, B be invertible complex matrices, and suppose for $0 < t < t_0 < 1$*

$$n_A(t) \le C_A L t^\alpha, \qquad n_B(t) \le C_B L t^\alpha, \qquad \alpha > 0.$$

Put $\delta = \|A - B\|_1$. For every $0 < r < t_0$,

$$\big|\log|\det A| - \log|\det B|\big| \le \frac{(C_A + C_B)L}{\alpha} r^\alpha + \frac{\delta}{r}. \tag{7}$$

The same bound holds for invertible identity-plus-trace-class operators when common finite orthogonal compressions converge in trace norm and have the stated uniform counting bounds. In particular, at the source interface (5)–(6)*,*

$$\big|\log|\det T| - \log|\det T_{SS}|\big| \le CL\xi^{1/3} \qquad (0 < \xi < \xi_0), \quad S = P\mathcal{X}. \tag{8}$$

Proof. Let $f_r(s) = \log\max(s, r)$. Its scalar Lipschitz constant on $[0, \infty)$ is $1/r$, so Lemma 2.1 bounds the difference of the two clipped singular-value sums by δ/r. For either matrix, Tonelli's theorem gives the exact identity

$$\sum_{s_i < r} \log(r/s_i) = \int_0^r \frac{n(t)}{t}\, dt \le \frac{CL}{\alpha} r^\alpha.$$

Since $|\det A| = \prod_i s_i(A)$, removing both clippings proves (7). In the stated infinite-dimensional setting, apply the finite result to the common compressions. The identity part has clipped logarithm zero, and continuity of the Fredholm determinant in trace norm, with nonzero limiting determinant, gives the claimed limit.

For the source take $\alpha = 1/2$, $\delta \le \xi L$ and $r = \xi^{2/3} < t_0$. Finite compressions of T inherit the source gap and count; choose an increasing sequence containing S. Their corresponding B's contain T_{SS} and identity blocks and satisfy the same count and trace-distance estimate. Thus the preceding limit argument applies. More generally r proportional to $(\delta/L)^{1/(\alpha+1)}$ gives exponent $\alpha/(\alpha+1)$. □

The source's Lemma `lem:03-schur`, lines 1020–1041, first compares T^*T and B^*B. The resulting bound is $CL(\sqrt{r} + \xi r^{-2})$ and yields $\xi^{1/5}$. Theorem 2.2 keeps the singular values themselves and replaces r^{-2} by r^{-1}. It is not a new general singular-value theorem. It is a quantitative improvement of this source's stated bound.

2.2 What does and does not improve

The source's Schur operator

$$T_e = T_{RR} - T_{RS}T_{SS}^{-1}T_{SR}$$

still has

$$\sum_{\lambda<1,\ \lambda\in\mathrm{spec}(\mathrm{Re}\,T_e)} -\log\lambda \le CL\xi^{1/3}(1+|\log\xi|). \tag{9}$$

We do not claim to remove this logarithm. Its proof uses the exact lift $Jy = (-T_{SS}^{-1}T_{SR}y, y)$:

$$J^*(\mathrm{Re}\,T)J = \mathrm{Re}\,T_e, \qquad \|Jy\| \ge \|y\|\,.$$

This proves the inherited gap and count by min–max. Splitting T_{SS}^{-1} at singular-value threshold a gives a rank $O(L\sqrt{a})$ part and a trace perturbation $O(\xi L(1+a^{-1}))$; clipping the negative logarithm and choosing $a = r = \xi^{2/3}$ gives (9). Consequently the combined two local losses can now be bounded by $CL\xi^{1/3}(1+|\log\xi|)$, instead of the previous dominant $CL\xi^{1/5}$. The limits remain $L\to\infty$ first and $\xi\downarrow 0$ afterwards. This supplies neither a rate in N for the cylinder exponent nor a measured computational speedup.

Sharpness in a larger abstract class. Let L be an integer, $\epsilon = L^{-2}$, and let A, B equal identity except for one diagonal entry $\epsilon, 2\epsilon$, respectively. For small t, both counts are at most $L\sqrt{t}$. Their distance is $\delta = L^{-2}$, hence $\xi = \delta/L = L^{-3}$, whereas the log-determinant difference is $\log 2$. Thus $L\xi^{1/3} = 1$, and no larger power of ξ gives a uniform bound for all operator pairs under these hypotheses. This does *not* prove optimality in the more restrictive class $B = T_{SS}\oplus I$ arising from the source's cores.

A numerical scale, not a runtime claim. If $\xi = 10^{-6}$, the old normalized power is $10^{-6/5}$, approximately 0.0631, and the new one is 10^{-2}. Unknown constants and the separate Schur loss prevent interpreting this comparison as a certified numerical error for a particular cylinder. The benefit is the improved asymptotic dependence on a certified physical trace-distance tolerance.

3 Integer-grid continuation with a near-unit loss

The source's interpolation module does not define a gas with a noninteger number of particles. It constructs a holomorphic approximant $\widehat{F}(\nu_+,\nu_-,M)$ to genuine integer-particle integrals. Its approximation includes the reference normalization $J_{\nu_+}J_{\nu_-}$. Finite Hermite expansions, finite Legendre projection kernels, and separate endpoint expansions are indispensable to that construction. The following result improves only the last transfer of an integer-grid bound to a nearby complex particle parameter.

Lemma 3.1 (Polynomial grid and complex extension)**.** *Let p be a complex polynomial in d variables of degree at most k in each variable, R a positive integer, and*

$$A_k = 2\sum_{j=1}^{k} j^2 = \frac{k(k+1)(2k+1)}{3}, \qquad \theta = \frac{dA_k}{2R} < 1.$$

If $|p(n)|\le b$ for every $n\in\mathbb{Z}^d\cap[-R,R]^d$, then

$$|p(z)| \le \frac{b}{1-\theta}\exp\left(k\sum_{\ell=1}^{d} g(z_\ell/R)\right), \tag{10}$$

where $g = 0$ on $[-1,1]$, and otherwise $g(\zeta) = \log|\zeta + \sqrt{\zeta^2-1}|$, with the branch for which the modulus is greater than one. In particular, $g(x+iy) \le 2|y|$ when $|x|, |y| \le 1/2$.

Proof. Put $S = \sup_{[-R,R]^d} |p|$. Holding all other variables real, expand the one-variable restriction in Chebyshev polynomials. Its constant coefficient has modulus at most S, and every other coefficient at most $2S$, by the cosine integral formula. Also $|T_j'(x)| \le j^2$ on $[-1,1]$, since $T_j'(\cos t) = j\sin(jt)/\sin t$ and $|\sin(jt)| \le j|\sin t|$. Thus each partial derivative on the cube has modulus at most $A_k S/R$. Join a maximizing point to its nearest integer grid point, coordinate by coordinate. Each displacement is at most $1/2$; the resulting bound $S \le b + \theta S$ gives $S \le b/(1-\theta)$.

For a one-variable polynomial q bounded by S on $[-1,1]$, the function

$$w^{-k} q((w + w^{-1})/2)$$

is analytic for $|w| > 1$, including infinity, and bounded by S on $|w| = 1$. The maximum principle gives $|q(\zeta)| \le S e^{k g(\zeta)}$. Apply this successively in each coordinate; this is the classical interval Bernstein–Walsh argument, proved here in the needed form. Finally $g(x+iy) = |\operatorname{Im} \arccos(x+iy)|$ in the indicated central rectangle. Along its vertical segment $|(\arccos)'(\zeta)| = |1-\zeta^2|^{-1/2} \le 2$; integration from the real axis proves the last bound. □

Theorem 3.2 (An explicit analytic transfer certificate)**.** *Let m be an integer and let F be holomorphic on a neighborhood of the closed polydisk $|\nu_\ell - m| \le a$, with $|F| \le H$. Suppose that actual integer data $F_{\mathrm{int}}(n)$ obey*

$$|F_{\mathrm{int}}(n)| \le B, \qquad |F(n) - F_{\mathrm{int}}(n)| \le e_{\mathrm{int}}$$

on the integer cube $|n_\ell - m| \le R = \lfloor a/3 \rfloor$. Let p be its Taylor polynomial truncated at degree k in each variable, and let e_{pol} bound $|F - p|$ on $|\nu_\ell - m| \le a/2$. If $R \ge 1$, $\theta = dA_k/(2R) < 1$, and the target belongs to that half polydisk, then

$$|F(\nu^*)| \le e_{\mathrm{pol}} + \frac{B + e_{\mathrm{int}} + e_{\mathrm{pol}}}{1-\theta} \exp\left(k \sum_{\ell=1}^{d} g((\nu_\ell^* - m)/R) \right). \tag{11}$$

For $d = 2$, one may take $e_{\mathrm{pol}} = 8H\,2^{-k}$.

Proof. Cauchy's coefficient bound gives $|c_{\mathbf{j}}| \le H a^{-|\mathbf{j}|}$. On the half polydisk, the discarded terms are bounded by

$$H \sum_{\max(j_1,j_2) > k} 2^{-j_1 - j_2} \le 8H\,2^{-k}$$

in dimension two. At every integer point, $|p| \le B + e_{\mathrm{int}} + e_{\mathrm{pol}}$. Apply Lemma 3.1 to $p(m + \cdot)$, and add the target approximation error. □

Corollary 3.3 (Application to the source's bidisk)**.** *Retain the source's fixed analytic and integer estimates:*

$$a = \sqrt{m}, \quad m = N/2, \quad |F| \le e^{L^C}, \quad e_{\mathrm{int}} \le e^{-L^b}, \quad B \ge e^{-L^{b'}}, \quad b > b', \quad |\nu_\ell^* - m| \le L^D.$$

All constants and exponents here are fixed before $L \to \infty$. For a polynomial choice of k making $8e^{L^C} 2^{-k} \le e^{-L^b}$,

$$|F(\nu^*)| \le B(1 + o(1)). \tag{12}$$

An explicit relative upper factor is the right side of (11) *divided by B.*

Proof. Now R grows as $e^{\gamma L/2}$. Hence $\theta = O(k^3/R) \to 0$. For large L the normalized target lies in the central rectangle of Lemma 3.1, so the exponential factor is at most $\exp(2kdL^D/R) = 1 + o(1)$. Both additive errors divided by B are at most $\exp(-L^b + L^{b'})$. Substitution proves the claim. □

The source's Lemma `lem:04-grid` bounds a complex target by a tensor Chebyshev coefficient sum and retains a factor $C(k+1)^2$, adequate for $Be^{o(L)}$. The maximum-principle step above removes that factor. The elementary derivative bound was deliberately kept explicit; the classical sharp Markov inequality can replace A_k by k^2, but is unnecessary for (12). This is an improved *transfer* estimate, not a claim that the source's other errors have a known rate.

A finite example. For $d = 2, k = 2, R = 100$, one has $A_k = 10, \theta = 1/10$. For target $(m+i, m-i)$, the certificate gives a multiplier at most $(10/9)e^{0.08}$, apart from the displayed approximation errors. Increasing R, with k and the displacement fixed, makes this factor tend to one. Ignoring the analytic norm H, or testing a fixed number of integer values while permitting arbitrary degree, would invalidate this inference: a polynomial can vanish at those values and still be arbitrarily large at the target.

4 Normalized counts: an explicit conditional law

Throughout this section the exact positive polynomial (1) and source limit (3) are hypotheses. Write $a_N = \log N$, $Y_N = K/a_N$, and

$$\Lambda_\chi(t) = \tau(\chi e^t) - \tau(\chi), \qquad t \in \mathbb{R}.$$

The finite identity

$$\frac{1}{a_N} \log \mathbb{E}_{N,\chi} e^{tK} = \frac{\log Z_N(\chi e^t) - \log Z_N(\chi)}{a_N} \longrightarrow \Lambda_\chi(t) \tag{13}$$

requires no uniform-in-t fugacity asymptotic. Each invocation uses a fixed positive fugacity.

Proposition 4.1 (Mean profile and concentration)**.** *The function $m(q) = q\tau'(q)$ is continuous and strictly increasing from zero to infinity on $q > 0$, and*

$$m(q) = \frac{4}{3\pi^2} \begin{cases} \eta \cot \eta, & q = 2\cos\eta, \quad 0 < \eta < \pi/2, \\ 1, & q = 2, \\ v \coth v, & q = 2\cosh v, \quad v > 0. \end{cases} \tag{14}$$

Under $\mathbb{P}_{N,\chi}$, $Y_N \to m(\chi)$ in probability and $\mathbb{E}_{N,\chi} Y_N \to m(\chi)$.

Proof. Differentiate (4). The signs are important: $d\eta/d\log q = -\cot\eta$ and $dv/d\log q = \coth v$. The derivative of $\eta\cot\eta$ is negative, since $\sin\eta\cos\eta < \eta$; the derivative of $v\coth v$ is positive, since $\sinh v \cosh v > v$. At zero angle both tend to one. Their endpoint limits establish the range of m.

For $h > 0$, exponential Markov gives

$$\limsup_N a_N^{-1} \log \mathbb{P}_{N,\chi}\{Y_N \geq m(\chi) + \epsilon\} \leq \Lambda_\chi(h) - h(m(\chi) + \epsilon) < 0$$

when h is sufficiently small and fixed. A negative h gives the lower tail. This proves concentration. For the mean, set $f_N(t) = a_N^{-1} \log Z_N(\chi e^t)$. Convexity and differentiability of these finite log-sums give

$$\frac{f_N(0) - f_N(-h)}{h} \leq f_N'(0) = \mathbb{E}_{N,\chi} Y_N \leq \frac{f_N(h) - f_N(0)}{h}.$$

First take $N \to \infty$, then $h \downarrow 0$. This justifies a first derivative, not differentiation twice. □

Theorem 4.2 (Full count LDP)**.** *For every fixed $\chi > 0$, Y_N satisfies a large-deviation principle on $\mathbb{R}$ with speed $\log N$ and good rate*

$$I_\chi(y) = \begin{cases} +\infty, & y < 0, \\ \tau(\chi), & y = 0, \\ y\log(q/\chi) - \tau(q) + \tau(\chi), & y > 0, \quad q = m^{-1}(y). \end{cases} \tag{15}$$

Equivalently $I_\chi(y) = \sup_{q>0}\{y\log(q/\chi) - \tau(q) + \tau(\chi)\}$. It vanishes exactly at $y = m(\chi)$.

Proof. The derivative of the expression being maximized with respect to $\log q$ is $y - m(q)$. Proposition 4.1 gives the unique maximum for $y > 0$. For $y = 0$ take $q \downarrow 0$; for $y < 0$, the same limit gives infinity. The convex conjugate is nonnegative (choose $q = \chi$). Strict increase of m gives the unique zero. The rate is lower semicontinuous, and $I_\chi(y) \geq ty - \Lambda_\chi(t)$ for every fixed $t > 0$; it therefore has compact level sets on $[0, \infty)$.

For completeness we give the classical tilting proof rather than claiming a new large-deviation principle of general scope [3]. Exponential Markov and (13) bound a small neighborhood of a point y by $-ty + \Lambda_\chi(t) + |t|\epsilon$ on the logarithmic scale. Choose t approximating the conjugate at each point and take a finite cover to obtain the upper bound for compact sets. Moreover

$$\limsup_N a_N^{-1} \log \mathbb{P}_{N,\chi}\{Y_N \geq R\} \leq -tR + \Lambda_\chi(t).$$

Letting $R \to \infty$ proves exponential tightness and extends the upper bound to closed sets. Negative values have zero probability.

For $y > 0$, fix $q = m^{-1}(y)$ and $t = \log(q/\chi)$. The exact likelihood ratio on counts is

$$\frac{d\mathbb{P}_{N,\chi}}{d\mathbb{P}_{N,q}}(K) = e^{-tK}\frac{Z_N(q)}{Z_N(\chi)}.$$

On $|Y_N - y| < \epsilon$, it is bounded below by $\exp\{a_N[-ty - |t|\epsilon + (\log Z_N(q) - \log Z_N(\chi))/a_N]\}$. The tilted probability of this event tends to one by Proposition 4.1. This proves its lower exponential bound $-I_\chi(y) - |t|\epsilon$; let $\epsilon \downarrow 0$. At zero use the actual empty family: $\mathbb{P}_{N,\chi}\{K = 0\} = 1/Z_N(\chi)$, whose exponent is $-\tau(\chi)$. Taking any point of an open set proves the full open-set lower bound, including sets containing zero. □

4.1 Explicit values, planar scaling, and limitations

At $\chi = 2$, the typical cylinder count is $4\log N/(3\pi^2)$, whereas the empty-family probability has exponent $-1/6$. At $\chi = 1$, $\eta = \pi/3$, giving

$$\tau(1) = \frac{5}{54}, \qquad m(1) = \frac{4}{9\pi\sqrt{3}}.$$

For $y > 4/(3\pi^2)$, let $v > 0$ solve $y = 4v\coth v/(3\pi^2)$; then

$$I_\chi(y) = y\log(2\cosh v/\chi) - \frac{1}{6} - \frac{2v^2}{3\pi^2} + \tau(\chi).$$

In the lower branch, replace $2\cosh v$ by $2\cos\eta$ and replace $-2v^2/(3\pi^2)$ by $+2\eta^2/(3\pi^2)$. As $y \to \infty$, $v = 3\pi^2 y/4 + o(1)$ and the errors are exponentially small times a polynomial in y. Hence

$$I_\chi(y) = \frac{3\pi^2}{8}y^2 - y\log\chi + \tau(\chi) - \frac{1}{6} + o(1). \tag{16}$$

This is an expansion of the limiting rate, in the order $N \to \infty$ then $y \to \infty$. It is not a uniform estimate for a growing $k/\log N$.

The source's separate topology and pressure estimates yield

$$P_q(c_0N)^2 \le Z_N(q) \le C_{q,c_0} P_q(c_0N)^2, \qquad \frac{\log P_q(r)}{\log r} \longrightarrow \frac{\tau(q)}{2}.$$

For the normalized number of polygons in one planar nest with every diameter at most r, the same proof therefore gives typical count $\frac{1}{2}m(\chi)\log r$, speed $\log r$, and rate

$$I_\chi^{\text{plane}}(y) = \frac{1}{2} I_\chi(2y). \tag{17}$$

This requires the comparison at *every fixed positive* q. A comparison only at $q = 2$ would not supply the tilted laws. Independence of the two small embedded nests in the comparison does not assert that the full cylinder ensemble is exactly their product.

Remark 4.3 (No variance or CLT from pressure alone). Multiply any positive partition polynomial $Z_N(q)$ with constant coefficient one by $1 + q^{b_N}$, where $b_N = \lfloor (\log N)^{3/4} \rfloor$. For every fixed $q > 0$, $\log(1 + q^{b_N})/\log N \to 0$, so all limiting pressures stay the same. At $q = 1$, the new count is the independent sum of the old count and a variable equal to $0, b_N$ with probabilities $1/2, 1/2$. Its variance is at least $b_N^2/4$, much larger than $\log N$. This is an abstract positive-polynomial counterexample, not a counterexample to the actual honeycomb model. It shows why the above argument does not prove a CLT or convergence of $\mathrm{Var}(K)/\log N$.

5 Ram comparison and preserved source mechanisms

5.1 Fresh original proof passages

The original local-coordinate and continuation arguments were read afresh in the confidential corpus, not inferred from catalogue summaries. Their roles are limited and explicit:

1. In TR117 [8], lines 208–315 establish the relation between the global integral and local coordinates with the required integrability; lines 715–785 discuss finite residual/inversion conditions. A forward polynomial representation is not automatically a stable inverse, and an infinite expansion needs more than a collection of finite identities. Here this distinction separates the finite core from the source's uniform trace and counting bounds.
2. In TR018 [9], lines 1155–1244 and 1713–1840 treat common analytic domains, continuation, interpolation and stability. Branches, cycles and a dominating integrable bound remain part of the data. Our grid theorem is a finite, explicit instance of this domain-aware logic: its input is an already constructed holomorphic function with a norm bound and controlled integer errors.
3. The original local-moment chapter [10], lines 1–91, gives Taylor expansion with an integral remainder, retaining raw moments and the required regularity. It supports keeping approximation remainders explicit, not replacing the actual gas by its first two moments. The Cauchy remainder in Theorem 3.2 is tailored to the source's holomorphic bidisk rather than asserting a new general Taylor identity.

These original methods motivate the interfaces and the error accounting. They do not prove the honeycomb transfer identities, the compensated gas inequalities, or the source's pressure asymptotics. In particular, the complex particle weights cannot be converted into a positive covariance inner product by naming them moments. The Hilbert norm used in Section 2 is the source's positive reference norm; the interaction itself remains complex.

5.2 Signs, topology, and ordered limits that survive the refinement

The all-count bound in the source reads, for sufficiently large even N and all $v \geq 0$,

$$Z_N(2\cosh v) \leq \exp\{C \log N(1 + v^2)\}.$$

Its consequent $w_N(k) \leq \exp\{C \log N - ck^2/\log N\}$ is not a probability statement until the fugacity and partition normalization are restored. Our rate calculation uses the fixed-fugacity equality instead; the uniform all-count bound is not differentiated to obtain moments.

The source's complex sectors have an imbalance phase $e^{i(\eta+\pi f)M}$. Far fused pairs count as two elementary particles but carry one real integration coordinate; their factorials and activities are not those of two independent particles. The finite coefficient extraction and residue cancellation are kept unchanged. Similarly, the empirical variable is centered by the prescribed integer reference density, not silently by its expectation. Transport in the fluctuation argument is a finite particle-chain deformation, not a claim that a nonlinear map is globally holomorphic and injective. Our operator modulus estimate says nothing about the argument of its complex determinant.

For planar transfer, the annular pressure counts essential polygons modulo horizontal translations and bounds both total weight and the first horizontal-span moment. The open-strip pressure instead counts planar shapes modulo all triangular translations; one transfer step advances two columns. Its expansion $Q_N = c_1 N + c_2 + O(N^{-1})$ yields a diameter tail $O(R^{-2})$ by positive differences at two widths. Both are needed to remove essential and large contractible polygons at bounded multiplicative cost. No finite determinant estimate replaces these topology and positivity arguments.

All fixed smoothing, buffer, core and contour parameters precede the large-L limit in the source. In particular, $\lim_{\xi\downarrow 0} \limsup_{L\to\infty} L^{-1}\text{error} = 0$ does not give an $o(L)$ error at a fixed ξ. We found no correction needed to the signs or normalizations used in the selected deductions. The changes in this report are refinements, not a purported repair or full validation of the source.

5.3 Advantages, costs, and further work

The determinant refinement uses the same trace-distance certificate and small-value count and avoids an extra inverse power of the clipping threshold. Its cost is dependence on genuine infinite-dimensional counting control; finite probes alone cannot establish that control. The grid estimate is near-isometric on the source's exponentially large integer box, but requires both the large analytic domain and a bound on the full function there. It cannot extrapolate from sparse data with no complexity constraint.

The count law makes the role of fugacity operational: it gives a typical logarithmic count, a parametric rate for rare counts, and an explicit empty-family exponent. Its cost is dependence on the all-positive-fugacity source theorem. The planar factor $1/2$ rests on a separate geometric comparison. None of these statements proves convergence of loop shapes to a conformal loop law.

Useful next questions are to quantify the smooth-core rank as a function of ξ, sharpen the remaining Schur negative-logarithm loss, and retain explicit constants through the particle transport. A count CLT would require substantially stronger control, for example a uniform analytic expansion of log partition functions on a complex neighborhood with sufficiently small differentiated remainder. The positive-polynomial counterexample shows why a pointwise pressure limit alone is inadequate. No such additional input is claimed here.

6 Public context and evidence scope

Public primary pages were checked on October 9, 2026. Cardy's annulus paper [4] explicitly presents a Coulomb-gas proposal for a scaling partition function; Gamsa–Cardy [5] uses the proposed CFT relation to study twist observables. Neither abstract supplies an independent theorem about this discrete cylinder partition. Duminil-Copin–Smirnov [6] proves the honeycomb connective constant; this is not the cylinder all-fugacity exponent. The current version of Ang–Remy–Sun [7], dated February 15, 2025, describes rigorous annulus-modulus formulae for continuum random geometry, including the SLE loop partition. Its model and conclusions do not automatically identify the present lattice limit. These refreshed records establish context, not priority for our local inequalities or a blanket current resolution of lattice scaling. Only the cited abstracts/version records and publisher metadata were read online; their full proofs were not independently audited here.

The local source reading covers all 11 TeX files and the README: 6,188 lines in bounded literal displays, including all nine scientific modules. One truncated cylinder display was replaced by two untruncated displays before counting that interval. Fresh private proof coverage is 488 lines across the three originals listed above, not their complete files or included subchapters. The read ledger contains exact displayed intervals, unread complements, and hashes captured before and after the reads. The source identity comparison is snapshot provenance, not proof validation.

The owned executable checks supplement, rather than replace, the proofs: 1,250 parameter cases and 5,944 assertions passed. They use exact rational arithmetic for singular-value fixtures, clipped-log enclosures, polynomial/Joukowski identities, and normalized tilt identities. Their separate receipt distinguishes parameter cases from assertions. No source checker or parent's suite is replayed or counted. The exact suite and final build/visual receipts accompany this report.

Assessment. Applicability remains L3. C3 is assigned to the demonstrated $\xi^{1/3}$ finite-core refinement, the explicit near-unit grid transfer and their source-specific proof interfaces. The normalized count analysis is a useful conditional application of classical convexity and exponential tilting, not a newly established lattice exponent. The finite tests, full-source reading, and private provenance are evidence of scope and reproducibility, not independent reasons to raise the contribution score.

References

[1] OpenAI, *Cylinder loop weights and planar nesting*, September 26, 2026, captured source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public manuscript. The strip-crossing companion is an external input, not independently audited here.

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[7] M. Ang, G. Remy and X. Sun, *The moduli of annuli in random conformal geometry*, version 3, February 15, 2025. arXiv:2203.12398v3.

[8] M. Subbarao, original confidential Ram Foundation manuscript, `TR-rev-1-old-tr117.tex`, local directory `RamFoundationTRs/TR-rev-1-old-tr117`. Fresh passages: 208–315 and 715–785. No online disclosure.

[9] M. Subbarao, original confidential Ram Foundation manuscript, `TR-rev-1-old-tr018.tex`, local directory `RamFoundationTRs/TR-rev-1-old-tr018`. Fresh passages: 1155–1244 and 1713–1840.

[10] M. Subbarao, original confidential local-moment chapter, `ch03-local-moments-rev2.tex`, in `RamCalBooks`, volume `volume1-rev2`, chapters directory. Fresh passage: 1–91; other included chapters were not read.